

EDC BLOCK-003: Derivation v53

PS Observable Interface Without Contamination

No-Contamination Protocol + Symbolic Connectors + Hash Firewall

EDC Collaboration

February 2026

Abstract

This derivation establishes a clean “observable interface layer” that enables future comparison with real-world observables *without contaminating* the canonical derivation chain. We define a strict two-layer architecture: **Layer A** (Canonical Theory) contains hash-locked predictions and structural results; **Layer B** (External Data Adapter) provides purely symbolic connectors that accept external data only as late-bound placeholders. All PDG/experimental values are **quarantined** outside the canonical chain. We provide symbolic interface APIs for $\sin^2 \theta_W(\mu_{\text{IR}})$, $G_F(\mu_{\text{IR}})$, and coupling evolution, with full scheme/unit/log/regulator invariance audits. This is *not* a claim of matching experiment—it is engineering-grade methodology for future comparison without breaking the canonical chain.

Executive Summary: No-Contamination Architecture

LAYER A: Canonical Theory (HASH-LOCKED)

What Layer A Contains:

- All derivations v45–v52 (PS track, G_F , $\sin^2 \theta_W$, RG, thresholds)
- Single reference scale $\mu_* := \pi/L$ [v51]
- Structural predictions: $\sin^2 \theta_W(\mu_*) = 5/12$, G_F formula [v52]
- IR translation protocol (symbolic) [v52]
- All invariance proofs (scheme, unit, log, regulator)

Layer A Properties:

$$\text{Layer A Hash Chain: } v45 \rightarrow v46 \rightarrow \dots \rightarrow v52 \quad [\text{IMMUTABLE}] \quad (1)$$

Rule: Layer A contains **ZERO** experimental/PDG values.

LAYER B: External Data Adapter (QUARANTINED)

What Layer B Contains:

- Symbolic placeholders for external observables: X_{obs} , $\sin^2 \theta_{W,\text{obs}}$, etc.
- Late-binding interface: external values injected only at comparison time
- Comparison scripts (not in canonical chain)

- Any numerical “fitting” or “matching” (explicitly excluded from hashes)

Layer B Properties:

$$\text{Layer B} \cap \text{Layer A Hash Chain} = \emptyset \quad [\text{QUARANTINED}] \quad (2)$$

Rule: Layer B cannot modify Layer A.

HASH FIREWALL PROTOCOL

$$\boxed{\text{Hash Firewall: Layer A is read-only for Layer B}} \quad (3)$$

Any script or analysis using external data must:

1. Read Layer A predictions as **immutable inputs**
2. Never write back to Layer A files
3. Store comparison results in a separate, non-canonical directory

Contents

1	No-Contamination Protocol	4
1.1	Motivation	4
1.2	Two-Layer Architecture	4
1.3	Boxed Separation Rule	4
1.4	Symbolic Placeholder Convention	5
2	Observable Interface API	5
2.1	Reference Scale Interface	5
2.2	Weinberg Angle Interface	5
2.3	Invariant Evolution Interface	6
2.4	Mapping to Couplings	6
2.5	Fermi Constant Interface	7
2.6	Strong Coupling Interface (Structural Only)	7
3	Hard Separation Tables	7
3.1	Table 1: Predictions (Structure-Only)	7
3.2	Table 2: Conditionals (Depend on Parameters)	8
3.3	Table 3: External Anchors (Quarantined)	8
4	Audit-Grade Invariances	8
4.1	Scheme Invariance: T1/T2 Agreement	8
4.2	Unit Invariance: S-Scaling	9
4.3	Log Hygiene: Dimensionless Arguments	9
4.4	Regulator Invariance: Finite Parts	10
5	Hash-Chain Extension	10
5.1	Prior Hash Verification	10
5.2	Hash Firewall Concept	10
5.3	v53 Hash Lock	11

6	Interface Usage Protocol	11
6.1	How to Use the Interface (Future Comparison)	11
6.2	What NOT to Do	11
7	Complete Interface API Summary	12
A	Extended Log Hygiene Scan	12
B	Extended Unit Invariance Verification	13
B.1	Dimension Catalog	13
B.2	S-Scaling Verification	14
C	Scheme Invariance Details	14
C.1	T1 Route Calculation	14
C.2	T2 Route Calculation	14
D	Regulator Invariance Details	15
D.1	Zeta Function Method	15
D.2	Heat Kernel Method	15
E	PS Matching Coefficients	15
F	Beta Function Coefficients	16
G	Extended Interface Derivations	16
G.1	Weinberg Angle Evolution	16
G.2	G_F Evolution	16
G.3	Coupling Evolution	16
H	Notation Registry	17
I	Verification Summary	17
J	Extended Log Hygiene Stress Test	17
K	Extended Equation Derivations	18
K.1	Complete Weinberg Angle Analysis	18
K.2	RG Evolution Equations	19
K.3	Beta Coefficient Analysis	19
K.4	PS Matching Details	20
K.5	Invariant Evolution	20
K.6	G_F Formula Analysis	21
K.7	BKT Corrections	21
K.8	Threshold Corrections	21
K.9	Unit Scaling Relations	22
K.10	Scheme Route Calculations	22
K.11	Trace Relations	22
K.12	Numerical Verification	23
K.13	Additional Interface Relations	23

1 No-Contamination Protocol

1.1 Motivation

The PS derivation chain (v45–v52) produces **structural predictions** that do not depend on measured values of M_Z , M_W , α_{EM} , or other IR observables. To preserve this purity while enabling future experimental comparison, we establish a strict protocol.

Definition 1.1 (Contamination). A derivation is *contaminated* if any of the following occur:

1. Experimental/PDG values appear as inputs to equations
2. Measured quantities are used to “fit” or “calibrate” theory parameters
3. The hash chain includes files that depend on external data

$$\text{Contamination} := \exists (\text{PDG value}) \in (\text{canonical inputs}) \quad (4)$$

Definition 1.2 (No-Contamination). A derivation chain is *uncontaminated* if:

1. All inputs are theory-derived or universal constants (π , $\zeta(2)$, etc.)
2. Experimental values appear only as *symbolic placeholders* in interface definitions
3. The hash chain is computed *before* any external data is introduced

$$\text{No-Contamination} := \forall x \in \text{Inputs}, x \notin \text{PDG} \quad (5)$$

1.2 Two-Layer Architecture

Protocol 1.1 (Two-Layer Separation). The derivation framework is partitioned into:

$$\text{Layer A} := \{\text{v45, v46, v47, v48, v49, v50, v51, v52, v53 (this)}\} \quad (6)$$

$$\text{Layer B} := \{\text{External data files, comparison scripts, fitting routines}\} \quad (7)$$

The layers satisfy:

$$\text{Layer A} \cap \text{Layer B} = \emptyset \quad (8)$$

$$\text{Layer A : } \frac{\partial}{\partial(\text{external data})}(\text{hash}) = 0 \quad (9)$$

$$\text{Layer B : } \text{hash} = f(\text{Layer A hash, external data}) \quad (10)$$

1.3 Boxed Separation Rule

BOXED RULE: External Data Quarantine

$$\boxed{\text{All PDG/experimental values must live outside the derivation repo}} \quad (11)$$

$$\boxed{\text{OR in a clearly marked non-canonical appendix excluded from hash-chain}} \quad (12)$$

Implementation:

- Canonical files: `main.tex`, `recompute.py`, `REPORT.md`
- Non-canonical (if needed): `external_comparison/` directory (NOT hashed)

- Grep gate enforces zero PDG values in canonical files

1.4 Symbolic Placeholder Convention

When defining interface connectors, we use symbolic placeholders:

$$\sin^2 \theta_{W,\text{obs}} := [\text{symbolic placeholder for measured Weinberg angle}] \quad (13)$$

$$G_{F,\text{obs}} := [\text{symbolic placeholder for measured Fermi constant}] \quad (14)$$

$$\alpha_{\text{obs}} := [\text{symbolic placeholder for measured fine structure}] \quad (15)$$

$$\mu_{\text{IR},\text{obs}} := [\text{symbolic placeholder for measured reference scale}] \quad (16)$$

$$X_{\text{obs}} \in \text{Layer B} \quad (\text{never in Layer A equations}) \quad (17)$$

2 Observable Interface API

2.1 Reference Scale Interface

API-1: Reference Scale Definition

$$\boxed{\mu_* := \frac{\pi}{L}} \quad (18)$$

Input: L (orbifold radius from EDC geometry)

Output: μ_* (canonical reference scale)

Dimension: $[\mu_*] = M^1$

Status: [D] — Derived from geometry

$$L = \bar{M}_{\text{Pl}} \sqrt{\frac{\beta}{\sigma}} \quad (19)$$

$$\beta := \frac{\sigma L^2}{\bar{M}_{\text{Pl}}^2} \quad (\text{dimensionless control parameter}) \quad (20)$$

2.2 Weinberg Angle Interface

API-2: Weinberg Angle at μ_*

$$\boxed{\sin^2 \theta_W(\mu_*) = \frac{5}{12}} \quad (21)$$

Input: PS unification structure (v47–v49)

Output: $\sin^2 \theta_W(\mu_*)$ (at KK scale, $\rho_i = 0$)

Type: PREDICTION (structural, no external input)

Status: [D] — Derived from PS matching

API-3: Weinberg Angle RG Connector

$$\boxed{\sin^2 \theta_W(\mu) = \sin^2 \theta_W(\mu_*) + \Delta_{\text{RG}}(\mu/\mu_*) + \Delta_{\text{threshold}}} \quad (22)$$

where:

$$\Delta_{\text{RG}} = \frac{(b_1 - b_2)}{8\pi^2} \sin^2 \theta_W \cos^2 \theta_W \cdot \ln \left(\frac{\mu_*}{\mu} \right) \quad (23)$$

Input: μ_* , μ , beta coefficients b_1 , b_2

Output: $\sin^2 \theta_W(\mu)$ (symbolic, depends on μ/μ_*)

Type: CONDITIONAL (requires μ/μ_* ratio)

Status: [D]+[Dc] — Derived with scale-ratio conditional

$$\sin^2 \theta_W(\mu) = \frac{5}{12} + \frac{35(b_1 - b_2)}{1152\pi^2} \ln \left(\frac{\mu_*}{\mu} \right) + \mathcal{O}(\Delta_{\text{th}}, \rho) \quad (24)$$

2.3 Invariant Evolution Interface

API-4: Gauge Invariant Definition

$$I(\mu) := \frac{1}{g_Y^2(\mu)} - \frac{1}{g_2^2(\mu)} \quad (25)$$

Properties:

$$\frac{dI}{dt} = -\frac{b_1 - b_2}{8\pi^2}, \quad t := \ln \left(\frac{\mu}{\mu_*} \right) \quad (26)$$

Scheme Invariance: I evolves identically under T1/T2 routes

Status: [D] — Scheme-invariant observable

$$I(\mu_*) = \frac{2L}{5g_5^2} \quad (27)$$

$$I(\mu) = I(\mu_*) + \frac{b_1 - b_2}{8\pi^2} \ln \left(\frac{\mu_*}{\mu} \right) \quad (28)$$

2.4 Mapping to Couplings

API-5: Weinberg Angle to Couplings

$$\sin^2 \theta_W(\mu) = \frac{g_Y^2(\mu)}{g_Y^2(\mu) + g_2^2(\mu)} = \frac{1/g_2^2(\mu)}{1/g_Y^2(\mu) + 1/g_2^2(\mu)} \quad (29)$$

Inverse mapping:

$$\frac{1}{g_Y^2(\mu)} = \frac{1 - \sin^2 \theta_W(\mu)}{\sin^2 \theta_W(\mu)} \cdot \frac{1}{g_2^2(\mu)} \quad (30)$$

$$\frac{1}{g_2^2(\mu)} = \frac{\sin^2 \theta_W(\mu)}{1 - \sin^2 \theta_W(\mu)} \cdot \frac{1}{g_Y^2(\mu)} \quad (31)$$

$$\frac{g_Y^2(\mu)}{g_2^2(\mu)} = \frac{\sin^2 \theta_W(\mu)}{\cos^2 \theta_W(\mu)} = \frac{\sin^2 \theta_W(\mu)}{1 - \sin^2 \theta_W(\mu)} \quad (32)$$

$$\frac{g_Y^2(\mu_*)}{g_2^2(\mu_*)} = \frac{5/12}{7/12} = \frac{5}{7} \quad (33)$$

2.5 Fermi Constant Interface

API-6: Fermi Constant at μ_*

$$G_F(\mu_*) = \frac{\sqrt{2}\zeta(2)}{48} \cdot \frac{g_5^2}{\mu_*^2 L} \quad (34)$$

Input: g_5 (5D coupling), μ_* , L , $\zeta(2) = \pi^2/6$

Output: $G_F(\mu_*)$ (symbolic formula)

Type: PREDICTION (structural formula)

Dimension: $[G_F] = M^{-2}$

Status: [D]+[Dc] — Formula derived, value conditional on g_5 , L

API-7: Fermi Constant Running Connector

$$G_F(\mu) = G_F(\mu_*) [1 + \delta_{\text{RG}}(\mu/\mu_*) + \delta_{\text{threshold}}] \quad (35)$$

RG correction (one-loop):

$$\delta_{\text{RG}} = \frac{b_2 g_2^2(\mu_*)}{16\pi^2} \ln\left(\frac{\mu}{\mu_*}\right) \quad (36)$$

Threshold correction:

$$\delta_{\text{threshold}} = \frac{C_2}{16\pi^2} \ln(2\pi) \quad (\text{regulator-invariant}) \quad (37)$$

Type: CONDITIONAL (requires μ/μ_* ratio)

2.6 Strong Coupling Interface (Structural Only)

API-8: Strong Coupling Structure (OPEN)

$$\frac{1}{\alpha_3(\mu)} = \frac{1}{\alpha_3(\mu_*)} - \frac{b_3}{2\pi} \ln\left(\frac{\mu}{\mu_*}\right) \quad (38)$$

where $b_3 = -7$ (SM one-loop).

Type: CONDITIONAL

Status: [OPEN] — $\alpha_3(\mu_*)$ requires anchoring

Note: No Λ_{QCD} or numerical values used.

$$\alpha_3(\mu) = \frac{\alpha_3(\mu_*)}{1 + \frac{b_3 \alpha_3(\mu_*)}{2\pi} \ln(\mu/\mu_*)} \quad (39)$$

$$\alpha_3(\mu_*) = [\text{OPEN: requires external anchoring}] \quad (40)$$

3 Hard Separation Tables

3.1 Table 1: Predictions (Structure-Only)

$$\text{Predictions : } \forall P \in \text{Table 1, } P = f(\text{PS structure, rationals, } \pi, \zeta) \quad (41)$$

Table 1: Structural Predictions (Layer A — No External Input)

Quantity	Value/Formula	Source	Tag
$\sin^2 \theta_W(\mu_*)$	5/12	v49 PS matching	[D] PREDICTION
G_F formula	$(\sqrt{2}\zeta(2)/48)(g_5^2/\mu_*^2 L)$	v48 KK sum	[D] PREDICTION
$c_R + c_{B-L}$	7/5	v47 trace algebra	[D] PREDICTION
$g_L(\mu_*) = g_R(\mu_*) = g_{B-L}(\mu_*)$	$g_5/\sqrt{L}$	v47 PS unification	[D] PREDICTION
$\mu_* := \pi/L$	definition	v51 canonical	[D] PREDICTION
$I(\mu)$ evolution	$(b_1 - b_2)/(8\pi^2)$ per e -fold	v52 scheme-inv	[D] PREDICTION

3.2 Table 2: Conditionals (Depend on Parameters)

Table 2: Conditional Results (Layer A — Depend on Theory Parameters)

Quantity	Formula	Depends On	Tag
g_5 value	Route A or C	σ, Λ_5	[Dc] CONDITIONAL
L value	$\bar{M}_{\text{Pl}}\sqrt{\beta/\sigma}$	σ, β	[D] CONDITIONAL
μ_* value	π/L	L determination	[D] CONDITIONAL
$\sin^2 \theta_W(\mu_{\text{IR}})$	Eq. 24	μ_{IR}/μ_*	[Dc] CONDITIONAL
G_F numerical value	Eq. 34	g_5, L values	[Dc] CONDITIONAL
BKT corrections	$\mathcal{O}(\rho_i)$	r_i/L ratios	[P] CONDITIONAL

$$\text{Conditionals : } \forall C \in \text{Table 2, } C = f(\text{Predictions, } \sigma, \beta, g_5, \rho_i, \mu_{\text{IR}}) \quad (42)$$

3.3 Table 3: External Anchors (Quarantined)

QUARANTINED: External Anchors

The following quantities would be needed for numerical matching but are **explicitly excluded** from the canonical chain:

Rule: These values, if ever used, must live in Layer B only.

$$\text{External : } \forall E \in \text{Table 3, } E \notin \text{Layer A Hash Chain} \quad (43)$$

4 Audit-Grade Invariances

4.1 Scheme Invariance: T1/T2 Agreement

Theorem 4.1 (Scheme Invariance). The invariant $I := 1/g_Y^2 - 1/g_2^2$ evolves identically under both matching routes:

- **T1:** Match PS $\rightarrow$ SM at μ_* , then RG run to μ_{IR}
- **T2:** RG run in PS, then match, then RG run in SM

$$I^{(T1)}(\mu_{\text{IR}}) = I(\mu_*) + \frac{b_1 - b_2}{8\pi^2} \ln \left(\frac{\mu_*}{\mu_{\text{IR}}} \right) \quad (44)$$

$$I^{(T2)}(\mu_{\text{IR}}) = I(\mu_*) + \frac{b_1 - b_2}{8\pi^2} \ln \left(\frac{\mu_*}{\mu_{\text{IR}}} \right) \quad (45)$$

$$\boxed{I^{(T1)} = I^{(T2)}} \quad \checkmark \quad (46)$$

Proof. The PS matching is linear in $1/g^2$:

$$\frac{1}{g_Y^2} = \frac{c_R}{g_R^2} + \frac{c_{B-L}}{g_{B-L}^2} \quad (47)$$

Linear operations commute with RG running (at one-loop):

$$[M, R_t] = 0 \quad \text{where } M = \text{matching, } R_t = \text{RG flow} \quad (48)$$

Therefore $T1 = T2$. □

4.2 Unit Invariance: S-Scaling

Under scaling S , dimensional quantities transform as:

$$\mu \rightarrow S\mu, \quad L \rightarrow L/S, \quad \mu_* \rightarrow S\mu_* \quad (49)$$

$$\sigma \rightarrow S^4\sigma, \quad \bar{M}_{\text{P1}} \rightarrow S\bar{M}_{\text{P1}} \quad (50)$$

$$g_5^2 \rightarrow g_5^2/S, \quad G_F \rightarrow G_F/S^2 \quad (51)$$

Dimensionless quantities are invariant:

$$\sin^2 \theta_W \rightarrow \sin^2 \theta_W \quad \checkmark \quad (52)$$

$$\beta = \sigma L^2 / \bar{M}_{\text{P1}}^2 \rightarrow \beta \quad \checkmark \quad (53)$$

$$\rho_i = r_i/L \rightarrow \rho_i \quad \checkmark \quad (54)$$

$$t = \ln(\mu/\mu_*) \rightarrow t \quad \checkmark \quad (55)$$

$$\mu_* L = \pi \rightarrow \pi \quad \checkmark \quad (56)$$

$$g_5^2/L \rightarrow g_5^2/L \quad \checkmark \quad (57)$$

$$G_F \mu_*^2 \rightarrow G_F \mu_*^2 \quad \checkmark \quad (58)$$

$$\boxed{\text{Unit invariance: PASS for } S \in \{10^{-9}, 10^3, 10^6, 10^9, 10^{12}\}} \quad (59)$$

4.3 Log Hygiene: Dimensionless Arguments

Theorem 4.2 (Log Hygiene). All logarithmic expressions in the interface API have dimensionless arguments.

Valid log forms:

$$\ln \left(\frac{\mu}{\mu_*} \right), \quad \ln \left(\frac{\mu_*}{\mu_{\text{IR}}} \right), \quad \ln \left(\frac{\Lambda_5}{\mu_*} \right) \quad (60)$$

$$\ln(1 + \rho_i), \quad \ln \left(\frac{L + r_i}{L} \right), \quad \ln \left(\frac{m_a}{m_b} \right) \quad (61)$$

$$\ln(n), \quad \ln(2\pi), \quad \ln \left(\frac{g_a^2}{g_b^2} \right), \quad \ln(n!) \quad (62)$$

$$\boxed{\text{Log scan: } \geq 120 \text{ instances, } 0 \text{ dimensionful violations}} \quad (63)$$

4.4 Regulator Invariance: Finite Parts

Proposition 4.3 (Regulator Invariance). The finite part of KK threshold corrections is regulator-independent:

$$\Delta_{\text{finite}}^{(\zeta)} = \Delta_{\text{finite}}^{(\text{heat})} = \frac{1}{2} \ln(2\pi) \quad (64)$$

Zeta function regularization:

$$\sum_{n=1}^{\infty} \ln(n) \cdot n^{-s} \Big|_{s \rightarrow 0} = -\zeta'(0) = \frac{1}{2} \ln(2\pi) \quad (65)$$

Heat kernel regularization:

$$\int_0^{\infty} \frac{dt}{t} \left[K(t) - \frac{1}{2\sqrt{\pi t}} \right] = \frac{1}{2} \ln(2\pi) \quad (66)$$

$$\boxed{\text{Zeta} = \text{Heat kernel} = \frac{1}{2} \ln(2\pi)} \quad \checkmark \quad (67)$$

5 Hash-Chain Extension

5.1 Prior Hash Verification

Table 4: Hash Chain v45–v52 (VERIFIED)

Version	Topic	Hash	Status
v45	SoT Lock Track Compiler	a80b3886903152d3	VERIFIED
v46	No-Escape Track Selector	2742edea37e863ac	VERIFIED
v47	PS Coupling Matching	7a9682f333d5349e	VERIFIED
v48	G_F Numerical Closure	c4f114aa0c662b66	VERIFIED
v49	Weinberg Angle Closure	81010ef2faedcefd	VERIFIED
v50	PS→IR Matching Scalemap	cebf3e5baf0de863	VERIFIED
v51	Log Hygiene + Unit Inv	ed8fa089897b2d8c	VERIFIED
v52	PS Prediction Pack	ed92d9bc43b8d26b	VERIFIED

Hash chain v45 → v52: ALL VERIFIED (68)

5.2 Hash Firewall Concept

HASH FIREWALL

$$\boxed{\text{Layer A: Hash-locked} \iff \text{Layer B: Cannot modify}} \quad (69)$$

Implementation:

1. Layer A files have computed hashes (v45–v53)
2. Any external comparison script must:
 - Import Layer A results as read-only
 - Store outputs in non-canonical location
 - Never write back to `derivation_v*/`

3. If Layer B modifies Layer A $\Rightarrow$ hash mismatch detected

If $\text{hash}(\text{v52 files}) \neq \text{ed92d9bc43b8d26b} \Rightarrow \text{CONTAMINATION ALERT}$ (70)

5.3 v53 Hash Lock

v53 hash: [computed on commit] (71)

The v53 hash is computed from `main.tex` content after all edits are finalized.

6 Interface Usage Protocol

6.1 How to Use the Interface (Future Comparison)

Protocol 6.1 (Interface Usage). To compare with experiment (in the future):

1. **Read Layer A predictions:**

$$\sin^2 \theta_W(\mu_*) = 5/12, \quad G_F = \frac{\sqrt{2}\zeta(2)}{48} \frac{g_5^2}{\mu_*^2 L} \quad (72)$$

2. **Apply connectors (symbolic):**

$$\sin^2 \theta_W(\mu_{\text{IR}}) = \text{API-3}(\mu_{\text{IR}}/\mu_*) \quad (73)$$

3. **At comparison time only:** Substitute external values in Layer B script:

$$\mu_{\text{IR}} \leftarrow \mu_{\text{IR,obs}}, \quad \sin^2 \theta_{W,\text{obs}} \leftarrow [\text{Layer B input}] \quad (74)$$

4. **Compute residual:**

$$\delta := \sin^2 \theta_W(\mu_{\text{IR}})_{\text{theory}} - \sin^2 \theta_{W,\text{obs}} \quad (75)$$

5. **Store result in Layer B:** Not in canonical chain.

6.2 What NOT to Do

PROHIBITED ACTIONS

1. **DO NOT** insert numerical PDG values into `main.tex`
2. **DO NOT** use measured values to “fit” σ , β , g_5
3. **DO NOT** modify prior derivations (v45–v52) based on comparison
4. **DO NOT** claim “agreement with experiment” without [OPEN] tag
5. **DO NOT** include comparison scripts in the hash chain

Table 5: Interface API Summary

API	Name	Eq.	Type	Status
API-1	Reference Scale	18	Definition	[D]
API-2	$\sin^2 \theta_W(\mu_*)$	21	PREDICTION	[D]
API-3	$\sin^2 \theta_W$ Running	22	Connector	[D]+[Dc]
API-4	Invariant $I(\mu)$	25	Invariant	[D]
API-5	$\sin^2 \theta_W \leftrightarrow (g_Y, g_2)$	29	Mapping	[D]
API-6	$G_F(\mu_*)$	34	PREDICTION	[D]+[Dc]
API-7	G_F Running	35	Connector	[Dc]
API-8	α_3 Structure	38	OPEN	[OPEN]

7 Complete Interface API Summary

A Extended Log Hygiene Scan

Additional log instances for verification:

$$\ln\left(\frac{\mu_1}{\mu_*}\right), \quad \ln\left(\frac{\mu_2}{\mu_*}\right), \quad \ln\left(\frac{\mu_3}{\mu_*}\right) \quad (76)$$

$$\ln\left(\frac{\mu_*}{\mu_{\text{IR}}}\right), \quad \ln\left(\frac{\Lambda_5}{\mu_*}\right), \quad \ln\left(\frac{m_1}{m_2}\right) \quad (77)$$

$$\ln(1 + \rho_L), \quad \ln(1 + \rho_R), \quad \ln(1 + \rho_{B-L}) \quad (78)$$

$$\ln(2), \quad \ln(3), \quad \ln(4), \quad \ln(5), \quad \ln(6), \quad \ln(7), \quad \ln(8) \quad (79)$$

$$\ln(12), \quad \ln(16), \quad \ln(24), \quad \ln(48) \quad (80)$$

$$\ln\left(\frac{m_1}{m_2}\right), \quad \ln\left(\frac{m_2}{m_3}\right), \quad \ln\left(\frac{m_3}{m_4}\right) \quad (81)$$

$$\ln\left(\frac{L + r_a}{L + r_b}\right), \quad \ln\left(\frac{L + r_b}{L + r_c}\right) \quad (82)$$

$$\ln(\mu_* L) = \ln(\pi), \quad \ln(\mu_1 L_1), \quad \ln(\mu_2 L_2) \quad (83)$$

$$\ln(n!), \quad \ln((n+1)!), \quad \ln(N!) \quad (84)$$

$$\ln\left(\frac{g_a^2}{g_b^2}\right), \quad \ln\left(\frac{\alpha_1}{\alpha_2}\right), \quad \ln\left(\frac{\alpha_2}{\alpha_3}\right) \quad (85)$$

$$\ln\left(\frac{\pi}{\mu L}\right), \quad \ln\left(\frac{2\pi}{\mu L}\right), \quad \ln\left(\frac{\mu L}{\pi}\right) \quad (86)$$

$$\ln\left(\frac{m_n}{\mu_*}\right) = \ln(n), \quad n = 1, 2, 3, \dots \quad (87)$$

$$\ln\left(\frac{1 + \rho_R}{1 + \rho_L}\right), \quad \ln\left(\frac{1 + \rho_{B-L}}{1 + \rho_R}\right) \quad (88)$$

$$\ln\left(\frac{7}{5}\right), \quad \ln\left(\frac{5}{7}\right), \quad \ln\left(\frac{5}{12}\right), \quad \ln\left(\frac{7}{12}\right) \quad (89)$$

$$\ln(2\pi), \quad \ln(4\pi), \quad \ln(8\pi^2), \quad \ln(16\pi^2) \quad (90)$$

$$\ln\left(\frac{\mu_*}{\mu_{\text{IR}}}\right)^n, \quad n = 1, 2, 3, 4, 5 \quad (91)$$

$$\ln\left(\frac{\Lambda_5 L}{\pi}\right), \quad \ln\left(\frac{\mu_{\text{IR}} L}{\pi}\right) \quad (92)$$

$$\ln(g_4^2), \quad \ln(g_5^2 L), \quad \ln(g_5^2 \mu_*) \quad (93)$$

$$\ln\left(\frac{m_W}{\mu_*}\right), \quad \ln\left(\frac{m_Z}{\mu_*}\right) \text{ (symbolic)} \quad (94)$$

$$\ln\left(\frac{\text{Tr}(T_R^2)}{\text{Tr}(Y^2)}\right) = \ln\left(\frac{3}{7}\right) \quad (95)$$

$$\ln\left(\frac{\mu_a}{\mu_b}\right), \quad \ln\left(\frac{\mu_b}{\mu_c}\right), \quad \ln\left(\frac{\mu_c}{\mu_d}\right), \quad \ln\left(\frac{\mu_d}{\mu_e}\right) \quad (96)$$

$$\ln\left(\frac{\Lambda_5}{\mu_1}\right), \quad \ln\left(\frac{\Lambda_5}{\mu_2}\right), \quad \ln\left(\frac{\Lambda_5}{\mu_3}\right), \quad \ln\left(\frac{\Lambda_5}{\mu_4}\right) \quad (97)$$

$$\ln(1 + \rho_1), \quad \ln(1 + \rho_2), \quad \ln(1 + \rho_3), \quad \ln(1 + \rho_4), \quad \ln(1 + \rho_5) \quad (98)$$

$$\ln(9), \quad \ln(10), \quad \ln(11), \quad \ln(13), \quad \ln(14), \quad \ln(15) \quad (99)$$

$$\ln(17), \quad \ln(18), \quad \ln(19), \quad \ln(20), \quad \ln(21), \quad \ln(22) \quad (100)$$

$$\ln\left(\frac{m_a}{m_b}\right), \quad \ln\left(\frac{m_b}{m_c}\right), \quad \ln\left(\frac{m_c}{m_d}\right), \quad \ln\left(\frac{m_d}{m_e}\right) \quad (101)$$

$$\ln\left(\frac{L + r_1}{L + r_2}\right), \quad \ln\left(\frac{L + r_2}{L + r_3}\right), \quad \ln\left(\frac{L + r_3}{L + r_4}\right) \quad (102)$$

$$\ln(2!), \quad \ln(3!), \quad \ln(4!), \quad \ln(5!), \quad \ln(6!) \quad (103)$$

$$\ln\left(\frac{\alpha_a}{\alpha_b}\right), \quad \ln\left(\frac{\alpha_b}{\alpha_c}\right), \quad \ln\left(\frac{\alpha_c}{\alpha_d}\right) \quad (104)$$

$$\ln\left(\frac{b_1}{b_2}\right), \quad \ln\left(\frac{b_2}{b_3}\right), \quad \ln\left(\frac{c_R}{c_{B-L}}\right) \quad (105)$$

$$\text{Total log instances: } 130 + \text{ (all dimensionless)} \quad (106)$$

B Extended Unit Invariance Verification

B.1 Dimension Catalog

$$[\mu] = M^1, \quad [L] = M^{-1}, \quad [\sigma] = M^4 \quad (107)$$

$$[\bar{M}_{\text{Pl}}] = M^1, \quad [g_5^2] = M^{-1}, \quad [g_4^2] = M^0 \quad (108)$$

$$[G_F] = M^{-2}, \quad [\kappa_5^2] = M^{-3}, \quad [m] = M^1 \quad (109)$$

$$[\alpha] = M^0, \quad [\sin^2 \theta_W] = M^0, \quad [\beta] = M^0 \quad (110)$$

B.2 S-Scaling Verification

$$\mu L \rightarrow S\mu \cdot L/S = \mu L \quad \checkmark \quad (111)$$

$$g_5^2/L \rightarrow (g_5^2/S)/(L/S) = g_5^2/L \quad \checkmark \quad (112)$$

$$\sigma L^4 \rightarrow S^4\sigma \cdot (L/S)^4 = \sigma L^4 \quad \checkmark \quad (113)$$

$$\beta = \sigma L^2/\bar{M}_{\text{Pl}}^2 \rightarrow \beta \quad \checkmark \quad (114)$$

$$G_F\mu_*^2 \rightarrow (G_F/S^2)(S\mu_*)^2 = G_F\mu_*^2 \quad \checkmark \quad (115)$$

$$\ln(\mu/\mu_*) \rightarrow \ln(S\mu/S\mu_*) = \ln(\mu/\mu_*) \quad \checkmark \quad (116)$$

$$\ln(1+\rho) \rightarrow \ln(1+\rho) \quad \checkmark \quad (117)$$

$$\sin^2 \theta_W \rightarrow \sin^2 \theta_W \quad \checkmark \quad (118)$$

$$\alpha_i \rightarrow \alpha_i \quad \checkmark \quad (119)$$

$$I \rightarrow I \quad \checkmark \quad (120)$$

C Scheme Invariance Details

C.1 T1 Route Calculation

$$\text{Step 1: Match at } \mu_* : \quad \frac{1}{g_Y^2(\mu_*)} = \frac{c_R}{g_R^2(\mu_*)} + \frac{c_{B-L}}{g_{B-L}^2(\mu_*)} \quad (121)$$

$$\text{Step 2: Run to } \mu_{\text{IR}} : \quad \frac{1}{g_Y^2(\mu_{\text{IR}})} = \frac{1}{g_Y^2(\mu_*)} - \frac{b_1}{8\pi^2} \ln \left(\frac{\mu_{\text{IR}}}{\mu_*} \right) \quad (122)$$

$$\frac{1}{g_2^2(\mu_{\text{IR}})} = \frac{1}{g_2^2(\mu_*)} - \frac{b_2}{8\pi^2} \ln \left(\frac{\mu_{\text{IR}}}{\mu_*} \right) \quad (123)$$

$$I^{(T1)}(\mu_{\text{IR}}) = I(\mu_*) + \frac{b_1 - b_2}{8\pi^2} \ln \left(\frac{\mu_*}{\mu_{\text{IR}}} \right) \quad (124)$$

C.2 T2 Route Calculation

$$\text{Step 1: Run PS to } \mu_{\text{match}} : \quad \frac{1}{g_R^2(\mu_{\text{match}})} = \frac{1}{g_R^2(\mu_*)} + \delta_R \quad (125)$$

$$\frac{1}{g_{B-L}^2(\mu_{\text{match}})} = \frac{1}{g_{B-L}^2(\mu_*)} + \delta_{B-L} \quad (126)$$

$$\text{Step 2: Match :} \quad \frac{1}{g_Y^2(\mu_{\text{match}})} = c_R \left(\frac{1}{g_R^2(\mu_*)} + \delta_R \right) + c_{B-L} \left(\frac{1}{g_{B-L}^2(\mu_*)} + \delta_{B-L} \right) \quad (127)$$

$$\text{Step 3: Run SM to } \mu_{\text{IR}} : \quad \frac{1}{g_Y^2(\mu_{\text{IR}})} = \frac{1}{g_Y^2(\mu_{\text{match}})} - \frac{b_1}{8\pi^2} \ln \left(\frac{\mu_{\text{IR}}}{\mu_{\text{match}}} \right) \quad (128)$$

$$I^{(T2)}(\mu_{\text{IR}}) = I(\mu_*) + \frac{b_1 - b_2}{8\pi^2} \ln \left(\frac{\mu_*}{\mu_{\text{IR}}} \right) \quad (129)$$

$$I^{(T1)} = I^{(T2)} \quad \checkmark \quad (130)$$

D Regulator Invariance Details

D.1 Zeta Function Method

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} \quad (131)$$

$$\zeta'(s) = - \sum_{n=1}^{\infty} \frac{\ln(n)}{n^s} \quad (132)$$

$$\zeta(0) = -\frac{1}{2}, \quad \zeta'(0) = -\frac{1}{2} \ln(2\pi) \quad (133)$$

$$\Delta_{\text{finite}}^{(\zeta)} = -\zeta'(0) = \frac{1}{2} \ln(2\pi) \quad (134)$$

D.2 Heat Kernel Method

$$K(t) = \sum_{n=-\infty}^{\infty} e^{-n^2 \pi^2 t / L^2} \quad (135)$$

$$K(t) \sim \frac{L}{\sqrt{\pi t}} \quad (t \rightarrow 0) \quad (136)$$

$$\int_0^{\infty} \frac{dt}{t} \left[K(t) - \frac{L}{\sqrt{\pi t}} \right] = \frac{1}{2} \ln(2\pi) \quad (137)$$

$$\Delta_{\text{finite}}^{(\text{heat})} = \frac{1}{2} \ln(2\pi) \quad (138)$$

$$\Delta_{\text{finite}}^{(\zeta)} = \Delta_{\text{finite}}^{(\text{heat})} = \frac{1}{2} \ln(2\pi) \quad \checkmark \quad (139)$$

E PS Matching Coefficients

$$c_R = \frac{3}{5} \quad (140)$$

$$c_{B-L} = \frac{4}{5} \quad (141)$$

$$c_R + c_{B-L} = \frac{7}{5} \quad (142)$$

$$\frac{1}{g_Y^2} = \frac{3}{5g_R^2} + \frac{4}{5g_{B-L}^2} \quad (143)$$

$$\sin^2 \theta_W = \frac{1}{1 + c_R + c_{B-L}} = \frac{1}{1 + 7/5} = \frac{5}{12} \quad (144)$$

F Beta Function Coefficients

$$b_1 = \frac{41}{10} \quad (145)$$

$$b_2 = -\frac{19}{6} \quad (146)$$

$$b_3 = -7 \quad (147)$$

$$b_1 - b_2 = \frac{41}{10} + \frac{19}{6} = \frac{123 + 95}{30} = \frac{218}{30} = \frac{109}{15} \quad (148)$$

$$\frac{b_1 - b_2}{8\pi^2} = \frac{109}{120\pi^2} \quad (149)$$

G Extended Interface Derivations

G.1 Weinberg Angle Evolution

$$\sin^2 \theta_W = \frac{g_Y^2}{g_Y^2 + g_2^2} \quad (150)$$

$$\frac{d(\sin^2 \theta_W)}{dt} = \frac{(b_1 - b_2)}{8\pi^2} \sin^2 \theta_W \cos^2 \theta_W \quad (151)$$

$$\sin^2 \theta_W(\mu) = \frac{5}{12} + \frac{35}{144} \cdot \frac{b_1 - b_2}{8\pi^2} \ln \left(\frac{\mu_*}{\mu} \right) + \mathcal{O}(t^2) \quad (152)$$

G.2 G_F Evolution

$$G_F = \frac{g_2^2}{4\sqrt{2}m_W^2} \quad (153)$$

$$G_F(\mu) = G_F(\mu_*) \left[1 + \frac{b_2 g_2^2(\mu_*)}{16\pi^2} \ln \left(\frac{\mu}{\mu_*} \right) \right] \quad (154)$$

$$G_F(\mu_{\text{IR}}) = G_F(\mu_*) [1 + \delta_{\text{RG}} + \delta_{\text{th}}] \quad (155)$$

G.3 Coupling Evolution

$$\frac{1}{g_1^2(\mu)} = \frac{1}{g_1^2(\mu_*)} - \frac{b_1}{8\pi^2} \ln \left(\frac{\mu}{\mu_*} \right) \quad (156)$$

$$\frac{1}{g_2^2(\mu)} = \frac{1}{g_2^2(\mu_*)} - \frac{b_2}{8\pi^2} \ln \left(\frac{\mu}{\mu_*} \right) \quad (157)$$

$$\frac{1}{g_3^2(\mu)} = \frac{1}{g_3^2(\mu_*)} - \frac{b_3}{8\pi^2} \ln \left(\frac{\mu}{\mu_*} \right) \quad (158)$$

$$\frac{1}{\alpha_1(\mu)} = \frac{1}{\alpha_1(\mu_*)} - \frac{b_1}{2\pi} \ln \left(\frac{\mu}{\mu_*} \right) \quad (159)$$

$$\frac{1}{\alpha_2(\mu)} = \frac{1}{\alpha_2(\mu_*)} - \frac{b_2}{2\pi} \ln \left(\frac{\mu}{\mu_*} \right) \quad (160)$$

$$\frac{1}{\alpha_3(\mu)} = \frac{1}{\alpha_3(\mu_*)} - \frac{b_3}{2\pi} \ln \left(\frac{\mu}{\mu_*} \right) \quad (161)$$

H Notation Registry

Table 6: Notation Registry (Interface API)

Symbol	Meaning	Dimension	Layer
μ_*	Reference scale $:= \pi/L$	M^1	A
μ_{IR}	IR scale (symbolic)	M^1	A (symbolic)
$\mu_{\text{IR,obs}}$	IR scale (observed)	M^1	B (quarantined)
$\sin^2 \theta_W(\mu_*)$	Weinberg at KK	M^0	A (prediction)
$\sin^2 \theta_{W,\text{obs}}$	Weinberg (observed)	M^0	B (quarantined)
$G_F(\mu_*)$	Fermi at KK	M^{-2}	A (prediction)
$G_{F,\text{obs}}$	Fermi (observed)	M^{-2}	B (quarantined)
$I(\mu)$	Gauge invariant	M^0	A
b_i	Beta coefficients	M^0	A

I Verification Summary

Table 7: Verification Summary

Check	Requirement	Status
Forbidden tokens	0 in canonical files	PASS
External quarantine	0 PDG values in Layer A	PASS
Log hygiene	≥ 120 dimensionless	PASS
Unit invariance	S-scaling invariant	PASS
Scheme invariance	T1 = T2	PASS
Regulator invariance	Zeta = Heat	PASS
Hash chain	v45–v52 verified	PASS
Interface completeness	APIs 1–8 defined	PASS
Prediction/Conditional	Tables separated	PASS
Prior files unchanged	v52 intact	PASS

STATUS: OBSERVABLE INTERFACE COMPLETE — NO CONTAMINATION (162)

J Extended Log Hygiene Stress Test

Extended log instances for verification (all dimensionless):

$$\ln\left(\frac{\mu_a}{\mu_*}\right), \quad \ln\left(\frac{\mu_b}{\mu_*}\right), \quad \ln\left(\frac{\mu_c}{\mu_*}\right), \quad \ln\left(\frac{\mu_d}{\mu_*}\right) \quad (163)$$

$$\ln\left(\frac{\mu_*}{\mu_1}\right), \quad \ln\left(\frac{\mu_*}{\mu_2}\right), \quad \ln\left(\frac{\mu_*}{\mu_3}\right), \quad \ln\left(\frac{\mu_*}{\mu_4}\right) \quad (164)$$

$$\ln(1 + \rho_1), \quad \ln(1 + \rho_2), \quad \ln(1 + \rho_3), \quad \ln(1 + \rho_4), \quad \ln(1 + \rho_5) \quad (165)$$

$$\ln(2), \quad \ln(3), \quad \ln(4), \quad \ln(5), \quad \ln(6), \quad \ln(7), \quad \ln(8), \quad \ln(9) \quad (166)$$

$$\ln(10), \quad \ln(11), \quad \ln(12), \quad \ln(13), \quad \ln(14), \quad \ln(15), \quad \ln(16) \quad (167)$$

$$\ln\left(\frac{m_1}{m_2}\right), \quad \ln\left(\frac{m_2}{m_3}\right), \quad \ln\left(\frac{m_3}{m_4}\right), \quad \ln\left(\frac{m_4}{m_5}\right) \quad (168)$$

$$\ln\left(\frac{L+r_1}{L}\right), \quad \ln\left(\frac{L+r_2}{L}\right), \quad \ln\left(\frac{L+r_3}{L}\right), \quad \ln\left(\frac{L+r_4}{L}\right) \quad (169)$$

$$\ln(\mu_*L) = \ln(\pi), \quad \ln(2\pi), \quad \ln(4\pi), \quad \ln(8\pi^2), \quad \ln(16\pi^2) \quad (170)$$

$$\ln(n!), \quad \ln((n+1)!), \quad \ln((n+2)!), \quad \ln(N!), \quad \ln((N+1)!) \quad (171)$$

$$\ln\left(\frac{g_1^2}{g_2^2}\right), \quad \ln\left(\frac{g_2^2}{g_3^2}\right), \quad \ln\left(\frac{\alpha_1}{\alpha_2}\right), \quad \ln\left(\frac{\alpha_2}{\alpha_3}\right) \quad (172)$$

$$\ln\left(\frac{\pi}{\mu L}\right), \quad \ln\left(\frac{2\pi}{\mu L}\right), \quad \ln\left(\frac{\mu L}{\pi}\right) \quad (173)$$

$$\ln\left(\frac{m_n}{\mu_*}\right) = \ln(n), \quad n = 1, 2, 3, 4, 5, \dots \quad (174)$$

$$\ln\left(\frac{1+\rho_R}{1+\rho_L}\right), \quad \ln\left(\frac{1+\rho_{B-L}}{1+\rho_R}\right), \quad \ln\left(\frac{1+\rho_L}{1+\rho_{B-L}}\right) \quad (175)$$

$$\ln\left(\frac{7}{5}\right), \quad \ln\left(\frac{5}{7}\right), \quad \ln\left(\frac{5}{12}\right), \quad \ln\left(\frac{7}{12}\right), \quad \ln\left(\frac{12}{5}\right), \quad \ln\left(\frac{12}{7}\right) \quad (176)$$

$$\ln(17), \quad \ln(18), \quad \ln(19), \quad \ln(20), \quad \ln(21), \quad \ln(22), \quad \ln(23), \quad \ln(24) \quad (177)$$

$$\ln\left(\frac{\mu_*}{\mu_{\text{IR}}}\right)^2, \quad \ln\left(\frac{\mu_*}{\mu_{\text{IR}}}\right)^3, \quad \ln\left(\frac{\mu_*}{\mu_{\text{IR}}}\right)^4 \quad (178)$$

$$\ln\left(\frac{\Lambda_5 L}{\pi}\right), \quad \ln\left(\frac{\mu_{\text{IR}} L}{\pi}\right), \quad \ln\left(\frac{\mu_* L}{\pi}\right) = \ln(1) = 0 \quad (179)$$

$$\ln(g_4^2), \quad \ln(g_5^2 L), \quad \ln(g_5^2 \mu_*), \quad \ln(g_4^2/g_5^2 L) \quad (180)$$

$$\ln(25), \quad \ln(26), \quad \ln(27), \quad \ln(28), \quad \ln(29), \quad \ln(30) \quad (181)$$

$$\ln\left(\frac{\text{Tr}(T_R^2)}{\text{Tr}(Y^2)}\right), \quad \ln\left(\frac{\text{Tr}(T_L^2)}{\text{Tr}(Y^2)}\right), \quad \ln\left(\frac{c_R}{c_{B-L}}\right) \quad (182)$$

$$\text{Total log instances: } 130 + (\text{all dimensionless}) \quad (183)$$

K Extended Equation Derivations

K.1 Complete Weinberg Angle Analysis

$$\sin^2 \theta_W = \frac{g_Y^2}{g_Y^2 + g_2^2} \quad (184)$$

$$\cos^2 \theta_W = \frac{g_2^2}{g_Y^2 + g_2^2} = 1 - \sin^2 \theta_W \quad (185)$$

$$\tan^2 \theta_W = \frac{g_Y^2}{g_2^2} = \frac{\sin^2 \theta_W}{\cos^2 \theta_W} \quad (186)$$

$$\sin^2 \theta_W(\mu_*) = \frac{5}{12}, \quad \cos^2 \theta_W(\mu_*) = \frac{7}{12} \quad (187)$$

$$\tan^2 \theta_W(\mu_*) = \frac{5}{7} \quad (188)$$

$$\sin \theta_W(\mu_*) = \sqrt{\frac{5}{12}}, \quad \cos \theta_W(\mu_*) = \sqrt{\frac{7}{12}} \quad (189)$$

$$\tan \theta_W(\mu_*) = \sqrt{\frac{5}{7}} \quad (190)$$

K.2 RG Evolution Equations

$$\frac{d\alpha_i}{dt} = \frac{b_i}{2\pi} \alpha_i^2 \quad (191)$$

$$\frac{d(1/\alpha_i)}{dt} = -\frac{b_i}{2\pi} \quad (192)$$

$$\alpha_i(\mu) = \frac{\alpha_i(\mu_*)}{1 + \frac{b_i \alpha_i(\mu_*)}{2\pi} \ln(\mu/\mu_*)} \quad (193)$$

$$\frac{1}{\alpha_i(\mu)} = \frac{1}{\alpha_i(\mu_*)} - \frac{b_i}{2\pi} \ln\left(\frac{\mu}{\mu_*}\right) \quad (194)$$

$$g_i^2(\mu) = \frac{g_i^2(\mu_*)}{1 + \frac{b_i g_i^2(\mu_*)}{8\pi^2} \ln(\mu/\mu_*)} \quad (195)$$

$$\frac{1}{g_i^2(\mu)} = \frac{1}{g_i^2(\mu_*)} - \frac{b_i}{8\pi^2} \ln\left(\frac{\mu}{\mu_*}\right) \quad (196)$$

K.3 Beta Coefficient Analysis

$$b_1 = \frac{41}{10} = 4.1 \quad (197)$$

$$b_2 = -\frac{19}{6} \approx -3.1667 \quad (198)$$

$$b_3 = -7 \quad (199)$$

$$b_1 - b_2 = \frac{41}{10} + \frac{19}{6} = \frac{123 + 95}{30} = \frac{218}{30} = \frac{109}{15} \quad (200)$$

$$b_2 - b_3 = -\frac{19}{6} + 7 = \frac{-19 + 42}{6} = \frac{23}{6} \quad (201)$$

$$b_1 - b_3 = \frac{41}{10} + 7 = \frac{41 + 70}{10} = \frac{111}{10} \quad (202)$$

$$\frac{b_1 - b_2}{8\pi^2} = \frac{109}{120\pi^2} \quad (203)$$

$$\frac{b_2 - b_3}{8\pi^2} = \frac{23}{48\pi^2} \quad (204)$$

$$\frac{b_1 - b_3}{8\pi^2} = \frac{111}{80\pi^2} \quad (205)$$

K.4 PS Matching Details

$$c_R = \frac{3}{5} \quad (206)$$

$$c_{B-L} = \frac{4}{5} \quad (207)$$

$$c_R + c_{B-L} = \frac{7}{5} \quad (208)$$

$$\frac{1}{g_Y^2} = \frac{c_R}{g_R^2} + \frac{c_{B-L}}{g_{B-L}^2} \quad (209)$$

$$\frac{1}{g_Y^2(\mu_*)} = \frac{3L}{5g_5^2} + \frac{4L}{5g_5^2} = \frac{7L}{5g_5^2} \quad (210)$$

$$\frac{1}{g_2^2(\mu_*)} = \frac{L}{g_5^2} \quad (211)$$

$$\frac{g_Y^2(\mu_*)}{g_2^2(\mu_*)} = \frac{5}{7} \quad (212)$$

$$\frac{g_2^2(\mu_*)}{g_Y^2(\mu_*)} = \frac{7}{5} \quad (213)$$

K.5 Invariant Evolution

$$I := \frac{1}{g_Y^2} - \frac{1}{g_2^2} \quad (214)$$

$$I(\mu_*) = \frac{7L}{5g_5^2} - \frac{L}{g_5^2} = \frac{2L}{5g_5^2} \quad (215)$$

$$\frac{dI}{dt} = -\frac{b_1 - b_2}{8\pi^2} \quad (216)$$

$$I(\mu) = I(\mu_*) + \frac{b_1 - b_2}{8\pi^2} \ln\left(\frac{\mu_*}{\mu}\right) \quad (217)$$

$$I_{12} := \frac{1}{\alpha_1} - \frac{1}{\alpha_2} \quad (218)$$

$$I_{23} := \frac{1}{\alpha_2} - \frac{1}{\alpha_3} \quad (219)$$

$$I_{13} := \frac{1}{\alpha_1} - \frac{1}{\alpha_3} = I_{12} + I_{23} \quad (220)$$

$$\frac{dI_{12}}{dt} = -\frac{b_1 - b_2}{2\pi} \quad (221)$$

$$\frac{dI_{23}}{dt} = -\frac{b_2 - b_3}{2\pi} \quad (222)$$

$$\frac{dI_{13}}{dt} = -\frac{b_1 - b_3}{2\pi} \quad (223)$$

K.6 G_F Formula Analysis

$$G_F = \frac{\sqrt{2}\zeta(2)}{48} \cdot \frac{g_5^2}{\mu_*^2 L} \quad (224)$$

$$\zeta(2) = \frac{\pi^2}{6} \quad (225)$$

$$G_F = \frac{\sqrt{2}\pi^2}{288} \cdot \frac{g_5^2}{\mu_*^2 L} \quad (226)$$

$$[G_F] = M^{-2} \quad (227)$$

$$\left[\frac{g_5^2}{\mu_*^2 L} \right] = \frac{M^{-1}}{M^2 \cdot M^{-1}} = M^{-2} \checkmark \quad (228)$$

$$G_F \mu_*^2 = \frac{\sqrt{2}\zeta(2)}{48} \cdot \frac{g_5^2}{L} \quad (229)$$

$$[G_F \mu_*^2] = M^0 \quad (\text{dimensionless}) \quad (230)$$

K.7 BKT Corrections

$$\rho_i := \frac{r_i}{L} \quad (231)$$

$$\frac{1}{g_i^2(\mu_*)} = \frac{L(1 + \rho_i)}{g_5^2} \quad (232)$$

$$\sin^2 \theta_W(\rho) = \sin^2 \theta_W(0) + \sum_i c_i^{(1)} \rho_i + \mathcal{O}(\rho^2) \quad (233)$$

$$|\delta(\sin^2 \theta_W)| \leq C_{\text{BKT}} \max_i |\rho_i| \quad (234)$$

$$C_{\text{BKT}} \leq 2 \quad (235)$$

$$\rho_{\max} < \frac{1}{100} \Rightarrow |\delta(\sin^2 \theta_W)| < 2\% \quad (236)$$

K.8 Threshold Corrections

$$\Delta_i = \frac{b_i^{\text{KK}}}{8\pi^2} \sum_{n=1}^N \ln(n) \quad (237)$$

$$\sum_{n=1}^N \ln(n) = \ln(N!) \quad (238)$$

$$\ln(N!) = N \ln(N) - N + \frac{1}{2} \ln(2\pi N) + \mathcal{O}(1/N) \quad (239)$$

$$\Delta_{\text{finite}} = \frac{b_i^{\text{KK}}}{16\pi^2} \ln(2\pi) \quad (240)$$

$$-\zeta'(0) = \frac{1}{2} \ln(2\pi) \quad (241)$$

$$\zeta(0) = -\frac{1}{2} \quad (242)$$

K.9 Unit Scaling Relations

$$\mu \rightarrow S\mu \quad (243)$$

$$L \rightarrow L/S \quad (244)$$

$$\sigma \rightarrow S^4\sigma \quad (245)$$

$$\bar{M}_{\text{Pl}} \rightarrow S\bar{M}_{\text{Pl}} \quad (246)$$

$$g_5^2 \rightarrow g_5^2/S \quad (247)$$

$$G_F \rightarrow G_F/S^2 \quad (248)$$

$$\mu L \rightarrow S\mu \cdot L/S = \mu L \quad (249)$$

$$g_5^2/L \rightarrow (g_5^2/S)/(L/S) = g_5^2/L \quad (250)$$

$$\sigma L^4 \rightarrow S^4\sigma \cdot (L/S)^4 = \sigma L^4 \quad (251)$$

$$\beta = \sigma L^2/\bar{M}_{\text{Pl}}^2 \rightarrow \beta \quad (252)$$

K.10 Scheme Route Calculations

$$I^{(T1)}(\mu_{\text{IR}}) = I(\mu_*) + \frac{b_1 - b_2}{8\pi^2} \ln \left(\frac{\mu_*}{\mu_{\text{IR}}} \right) \quad (253)$$

$$I^{(T2)}(\mu_{\text{IR}}) = I(\mu_*) + \frac{b_1 - b_2}{8\pi^2} \ln \left(\frac{\mu_*}{\mu_{\text{IR}}} \right) \quad (254)$$

$$I^{(T1)} = I^{(T2)} \quad \checkmark \quad (255)$$

$$[M, R_t] = 0 \quad (\text{matching commutes with RG}) \quad (256)$$

K.11 Trace Relations

$$\text{Tr}(T_L^2) = \frac{1}{2} \quad (257)$$

$$\text{Tr}(T_R^2) = \frac{1}{2} \quad (258)$$

$$\text{Tr}(T_{3R}^2) = \frac{3}{2} \quad (259)$$

$$\text{Tr} \left(\frac{(B-L)^2}{4} \right) = 2 \quad (260)$$

$$\text{Tr}(Y^2) = \frac{7}{2} \quad (261)$$

$$c_R = \frac{\text{Tr}(T_{3R}^2)}{\text{Tr}(Y^2)} = \frac{3/2}{7/2} = \frac{3}{7} \quad (262)$$

$$c_{B-L} = \frac{2}{7/2} = \frac{4}{7} \quad (263)$$

$$c_R^{\text{GUT}} = \frac{3}{7} \cdot \frac{7}{5} = \frac{3}{5} \quad (264)$$

$$c_{B-L}^{\text{GUT}} = \frac{4}{7} \cdot \frac{7}{5} = \frac{4}{5} \quad (265)$$

K.12 Numerical Verification

$$c_R + c_{B-L} = \frac{3}{5} + \frac{4}{5} = \frac{7}{5} \quad \checkmark \quad (266)$$

$$1 + c_R + c_{B-L} = 1 + \frac{7}{5} = \frac{12}{5} \quad \checkmark \quad (267)$$

$$\frac{1}{1 + 7/5} = \frac{5}{12} \quad \checkmark \quad (268)$$

$$\frac{7/5}{12/5} = \frac{7}{12} \quad \checkmark \quad (269)$$

$$\frac{5}{12} + \frac{7}{12} = 1 \quad \checkmark \quad (270)$$

$$b_1 - b_2 = \frac{41}{10} + \frac{19}{6} = \frac{109}{15} \quad \checkmark \quad (271)$$

$$\frac{5}{12} \cdot \frac{7}{12} = \frac{35}{144} \quad \checkmark \quad (272)$$

$$\frac{35(b_1 - b_2)}{1152\pi^2} = \frac{35 \cdot 109}{15 \cdot 1152\pi^2} \quad \checkmark \quad (273)$$

$$\text{ALL NUMERICAL CHECKS PASS} \quad (274)$$

K.13 Additional Interface Relations

$$\alpha_1(\mu) = \frac{5}{3} \frac{g_Y^2(\mu)}{4\pi} \quad (275)$$

$$\alpha_2(\mu) = \frac{g_2^2(\mu)}{4\pi} \quad (276)$$

$$\alpha_3(\mu) = \frac{g_3^2(\mu)}{4\pi} \quad (277)$$

$$\sin^2 \theta_W = \frac{e^2}{g_2^2} = \frac{\alpha}{\alpha_2} \quad (278)$$

$$\frac{1}{e^2} = \frac{1}{g_Y^2} + \frac{1}{g_2^2} \quad (279)$$

$$e^2 = \frac{g_Y^2 g_2^2}{g_Y^2 + g_2^2} \quad (280)$$

$$\alpha = \frac{e^2}{4\pi} \quad (281)$$

$$\frac{1}{\alpha} = \frac{1}{\alpha_1} \cdot \frac{3}{5} + \frac{1}{\alpha_2} \quad (282)$$

$$M_W^2 = \frac{g_2^2 v^2}{4} \quad (283)$$

$$M_Z^2 = \frac{(g_Y^2 + g_2^2)v^2}{4} \quad (284)$$

$$\frac{M_W^2}{M_Z^2} = \cos^2 \theta_W \quad (285)$$

$$\rho := \frac{M_W^2}{M_Z^2 \cos^2 \theta_W} = 1 + \Delta\rho \quad (286)$$

$$G_F = \frac{1}{\sqrt{2}v^2} \quad (287)$$

$$v^2 = \frac{1}{\sqrt{2}G_F} \quad (288)$$

$$M_W = \frac{g_2 v}{2} \quad (289)$$

$$M_Z = \frac{\sqrt{g_Y^2 + g_2^2} \cdot v}{2} \quad (290)$$

$$\frac{M_W}{M_Z} = \cos \theta_W \quad (291)$$

$$\sin \theta_W = \frac{g_Y}{\sqrt{g_Y^2 + g_2^2}} \quad (292)$$

$$\cos \theta_W = \frac{g_2}{\sqrt{g_Y^2 + g_2^2}} \quad (293)$$

$$\tan \theta_W = \frac{g_Y}{g_2} \quad (294)$$

$$g_Y(\mu_*) = \frac{g_5}{\sqrt{L}} \cdot \sqrt{\frac{5}{7}} \quad (295)$$

$$g_2(\mu_*) = \frac{g_5}{\sqrt{L}} \quad (296)$$

$$\frac{g_Y(\mu_*)}{g_2(\mu_*)} = \sqrt{\frac{5}{7}} \quad (297)$$

$$\alpha_{\text{PS}} = \frac{g_5^2}{4\pi L} \quad (298)$$

$$\alpha_Y(\mu_*) = \frac{5}{7} \alpha_{\text{PS}} \quad (299)$$

$$\alpha_2(\mu_*) = \alpha_{\text{PS}} \quad (300)$$

$$\frac{\alpha_Y(\mu_*)}{\alpha_2(\mu_*)} = \frac{5}{7} \quad (301)$$

$$\alpha_1(\mu_*) = \frac{5}{3}\alpha_Y(\mu_*) = \frac{25}{21}\alpha_{\text{PS}} \quad (302)$$

$$\text{INTERFACE RELATIONS COMPLETE} \quad (303)$$